Insurance Premium — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Insurance Premium (MKM-IP-001)

Tests run on **04 April 2026 at 14:48:33**.

Table 1: Test Results Summary — Insurance Premium (MKM-IP-001)

Total Tests	36
Passed	36
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Insurance Premium

Test Description	Acceptance Criteria	Result	Time
All factors applied	`abs(result - round(expected, 2)) < 0.01`	Pass	0.000s
Basic calculation	`result == 1250.0`	Pass	0.000s
Claims factor increases premium	`with_claims > base`	Pass	0.000s
Clamped to max	`result == MAX_PREMIUM`	Pass	0.000s
Clamped to min	`result == MIN_PREMIUM`	Pass	0.000s
Contents premium added	`with_contents == base + 500.0`	Pass	0.000s
High flood risk increases premium	`high_flood > low_flood`	Pass	0.000s
Returns float	`isinstance(result, float)`	Pass	0.000s
Security factor reduces premium	`with_security < without`	Pass	0.000s
100 years is heritage	`get_age_premium_band(1925, current_year=2025) == 'heritage'`	Pass	0.000s
10 years is modern	`get_age_premium_band(2015, current_year=2025) == 'modern'`	Pass	0.000s
30 years is established	`get_age_premium_band(1995, current_year=2025) == 'establi...`	Pass	0.000s
Default current year	`result in AGE_PREMIUM_FACTORS`	Pass	0.000s
Established	`get_age_premium_band(1960, current_year=2025) == 'establi...`	Pass	0.000s
Exactly 29 years modern	`get_age_premium_band(1996, current_year=2025) == 'modern'`	Pass	0.000s
Exactly 99 years established	`get_age_premium_band(1926, current_year=2025) == 'establi...`	Pass	0.000s
Exactly 9 years new build	`get_age_premium_band(2016, current_year=2025) == 'new_build'`	Pass	0.000s
Heritage old	`get_age_premium_band(1850, current_year=2025) == 'heritage'`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Modern	`get_age_premium_band(2000, current_year=2025) == 'modern'`	Pass	0.000s
New build	`get_age_premium_band(2020, current_year=2025) == 'new_build'`	Pass	0.000s
Exactly 100 third band	`(lo, hi) == AREA_PREMIUM_BANDS[2][1]`	Pass	0.000s
Exactly 50 second band	`(lo, hi) == AREA_PREMIUM_BANDS[1][1]`	Pass	0.000s
Just below 100 second band	`(lo, hi) == AREA_PREMIUM_BANDS[1][1]`	Pass	0.000s
Just below 50 first band	`(lo, hi) == AREA_PREMIUM_BANDS[0][1]`	Pass	0.000s
Large area fourth band	`(lo, hi) == AREA_PREMIUM_BANDS[3][1]`	Pass	0.000s
Medium area second band	`(lo, hi) == AREA_PREMIUM_BANDS[1][1]`	Pass	0.000s
Returns tuple	`isinstance(result, tuple)`; `len(result) == 2`	Pass	0.000s
Small area first band	`(lo, hi) == AREA_PREMIUM_BANDS[0][1]`	Pass	0.000s
Age premium factors all positive	`lo > 0`; `hi >= lo`	Pass	0.000s
Area bands are sorted	`thresholds == sorted(thresholds)`	Pass	0.000s
Base rate range	`lo > 0`; `hi > lo`	Pass	0.000s
Construction type factors present	`'Brick' in CONSTRUCTION_TYPE_FACTORS`; `'Timber Frame' in CONSTRUCTION_TYPE_FACTORS`	Pass	0.000s
Flood risk factors increase with risk	`very_high_hi > very_low_lo`	Pass	0.000s
Flood risk factors present	`set(FLOOD_RISK_PREMIUM_FACTORS.keys()) == expected`	Pass	0.000s
Min max premium bounds	`MIN_PREMIUM == 200`; `MAX_PREMIUM == 20_000`	Pass	0.000s
Property type factors present	`set(PROPERTY_TYPE_PREMIUM_FACTORS.keys()) == expected`	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
04-Apr-2026	1.0	Initial test suite (36 tests)	D. Kelly